

FINANCIAL MOVIE ANALYSIS REPORT

This report explores the financial and popular aspects of movies by analyzing datasets derived from **IMDb** and **TMDb**. The objective is to identify trends, profitable genres, production countries, and other key indicators that can assist a film production company in making informed investment decisions. The analysis includes the creation of a unified dataset, transformations, KPIs, and visualizations to communicate the findings effectively.

Data Sources

Three different datasets were merged:

1. **IMDb Titles** – Title metadata, genre, runtime, adult flag.
2. **IMDb Ratings** – Average user rating and vote count.
3. **TMDb Movies** – Financial metrics like budget, revenue, production info.

Data Preparation

- **Merged on tconst/id** to create a unified master dataset.
- Cleaned and transformed using **Power Query** in Excel.
- Extracted releaseYear from releaseDate.
- Created Return on Investment Column (ROI)
- Split multi-value columns like genres, productionCountries, productionCountry and Genre.
- Removed or filtered invalid/empty/zero values from budget, revenue, and tconst.

TMDB Data - Power Query Editor

File Home Transform Add Column View

Close & Load Refresh Preview Advanced Editor Manage Query

Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine Parameters Data source settings Data Sources New Source Recent Sources Enter Data New Query

Queries [4] ratings titles TMDB Data TMDB_Data_Processed

Table.ExpandTableColumn(#"Merged Queries1", "ratings", {"averageRating", "numVotes"}, {"averageRating", "numVotes"})

	id	title	status	releaseDate	revenue	runtime
1	tt0003471	Released	11/24/1913	1800000		
2	tt0004391	Released	4/25/1914	5773000		
3	tt0004545	Released	11/15/1914	87028		
4	tt0004972	Released	2/8/1915	11000000		
5	tt0005078	Released	12/13/1915	137365		
6	tt0005393	Released	1/4/1915	102224		
7	tt0006140	Released	12/15/1915	102437		
8	tt0006333	Released	12/24/1916	8000000		
9	tt0006568	Released	10/16/1916	1390000		
10	tt0006864	Released	9/4/1916	1750000		
11	tt0007038	Released	5/7/1916	102768		
12	tt0009105	Released	11/17/1918	3000		
13	tt0009369	Released	8/1/1918	8000000		
14	tt010323	Released	2/27/1920	8811		
15	tt010466	Released	8/26/1919	3600000		
16	tt011549	Released	9/17/1920	3000000		

16 COLUMNS. 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2:14 AM

Query Settings

PROPERTIES

Name

TMDB Data

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Merged Queries
- Expanded titles
- Merged Queries1
- Expanded ratings

N	O	P	Q	R	S	T	U	V
id	title	status	releaseDate	revenue	runtime	productionCompanies	productionCountries	
1913	315.7894737	genres.1	Crime	productionCompanies: Independent Moving Pictures	productionCountries: United States of America			
1913	315.7894737	genres.2	Drama	productionCompanies: Independent Moving Pictures	productionCountries: United States of America			
1914	5.12291029	genres.1	Fantasy	productionCompanies: Universal Film Manufacturing Company	productionCountries: United States of America			
1914	5.12291029	genres.2	Romance	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1914	5.12291029	genres.3	Western	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1915	110	genres.1	Adventure	productionCompanies: Epoch Film Co.	productionCountries: United States of America			
1915	110	genres.2	History	productionCompanies: Epoch Film Co.	productionCountries: United States of America			
1915	110	genres.3	War	productionCompanies: Epoch Film Co.	productionCountries: United States of America			
1915	7.935127953	genres.1	Drama	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1915	7.935127953	genres.2	Drama	productionCompanies: Paramount	productionCountries: United States of America			
1915	6.765320979	genres.1	Western	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1915	6.765320979	genres.2	Romance	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1915	4.558428266	genres.1	Drama	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1915	4.558428266	genres.2	Romance	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1916	40	genres.1	Adventure	productionCompanies: The Universal Film Mfg. Co.	productionCountries: Bahamas			
1916	40	genres.2	Drama	productionCompanies: The Universal Film Mfg. Co.	productionCountries: Bahamas			
1916	40	genres.3	Action	productionCompanies: The Universal Film Mfg. Co.	productionCountries: Bahamas			
1916	40	genres.4	Science Fiction	productionCompanies: The Universal Film Mfg. Co.	productionCountries: Bahamas			
1916	1.39	genres.1	Fantasy	productionCompanies: Fox Film Corporation	productionCountries: United States of America			
1916	1.39	genres.2	Drama	productionCompanies: Fox Film Corporation	productionCountries: United States of America			
1916	4.534771331	genres.1	Drama	productionCompanies: Triangle Film Corporation	productionCountries: United States of America			
1916	4.534771331	genres.2	History	productionCompanies: Wark Producing Corp.	productionCountries: United States of America			
1916	4.534771331	genres.3	History	productionCompanies: Triangle Film Corporation	productionCountries: United States of America			
1916	5.532597577	genres.1	Drama	productionCompanies: Jesse L. Lasky Feature Play Company	productionCountries: United States of America			
1918	1	genres.1	Fantasy	productionCompanies: Herbert M. Dawley Production	productionCountries: United States of America			
1918	1	genres.2	Animation	productionCompanies: Herbert M. Dawley Production	productionCountries: United States of America			
1918	32	genres.1	Comedy	productionCompanies: Mabel Normand Feature Film Company	productionCountries: United States of America			

TMDB_Data_Processed TMDB Data titles ratings PIVOT ANALYSIS

Queries & Connections

Queries

Connections

4 queries

- ratings 1,048,575 rows loaded.
- titles 1,048,575 rows loaded. 4 errors.
- TMDB Data 5,790 rows loaded.
- TMDB_Data_Processed 31,432 rows loaded.

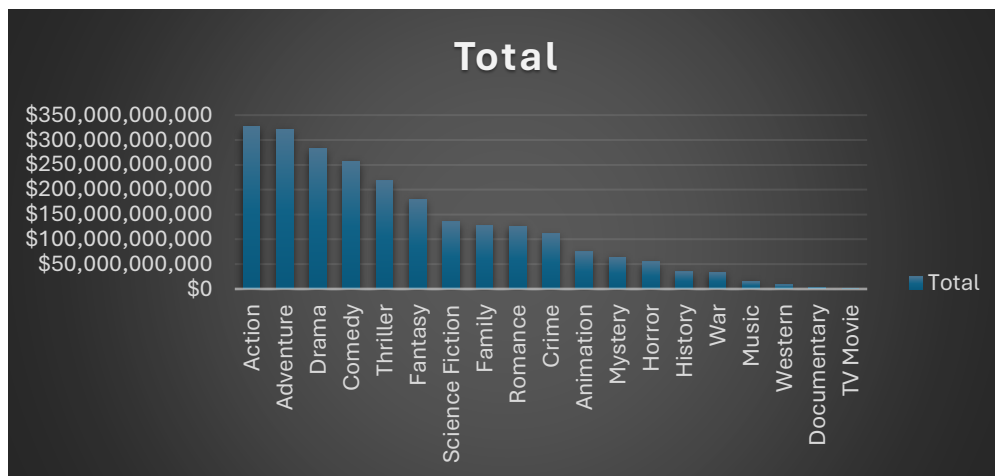
Key Performance Indicators (KPIs)

Metric	Value
Total Movies	5,791
Avg. Revenue	\$59.5 M
Avg. Budget	\$22.5 M
Avg. ROI	1,158%
Avg. Rating	6.43 / 10

Visual Analysis & Pivot Tables

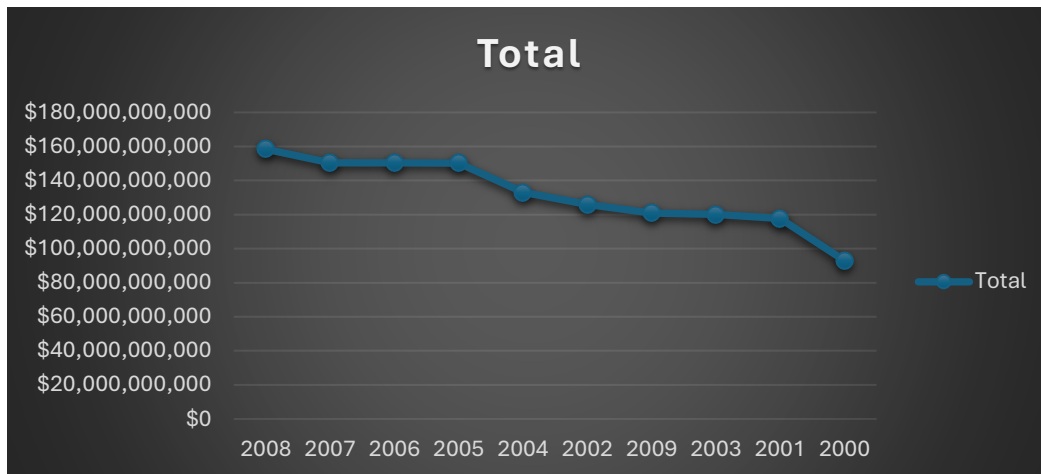
1. Revenue by Genre

- Genres like **Action**, **Adventure**, and **Animation** dominate in terms of total revenue.



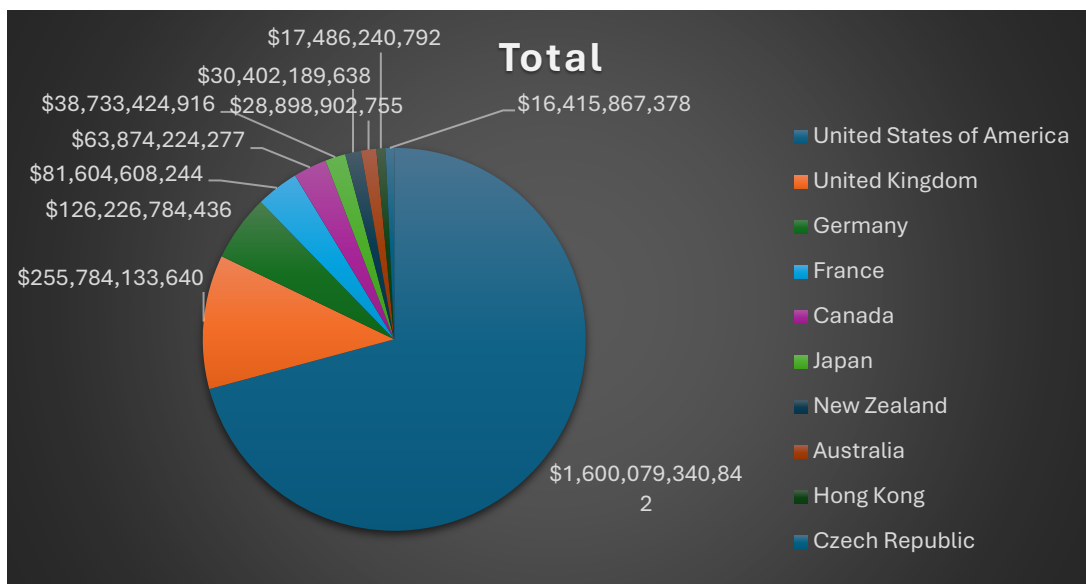
2. Average Rating by Genre & Release Year

- Helps in identifying which genres remain critically strong over time.
- Used releaseYear (extracted from releaseDate) and grouped genres.



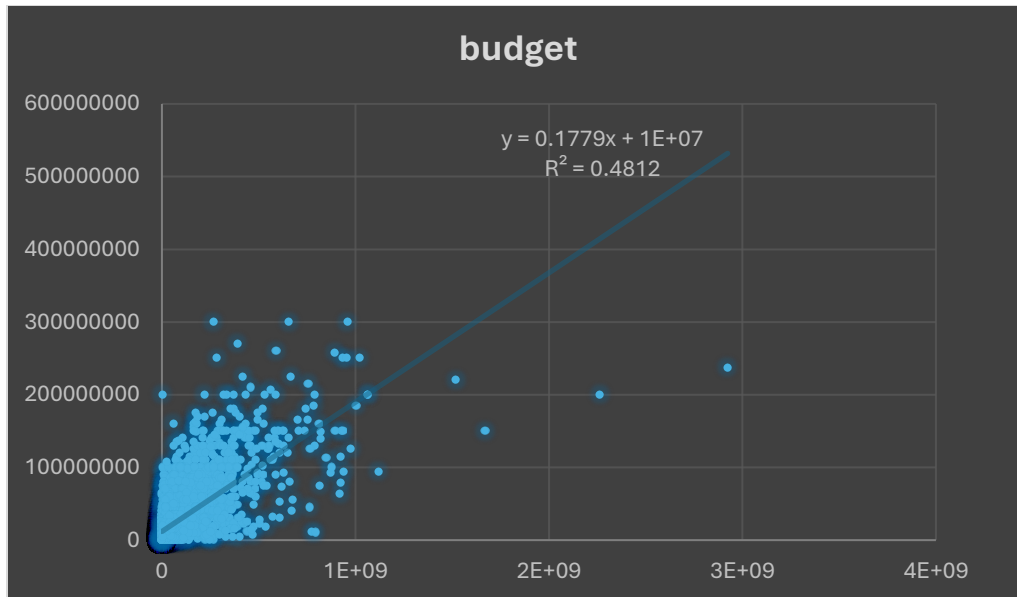
3. Revenue by Production Country

- USA, UK, and Canada were among the top grossing countries.
- Unpivoted productionCountries to separate multiple entries.



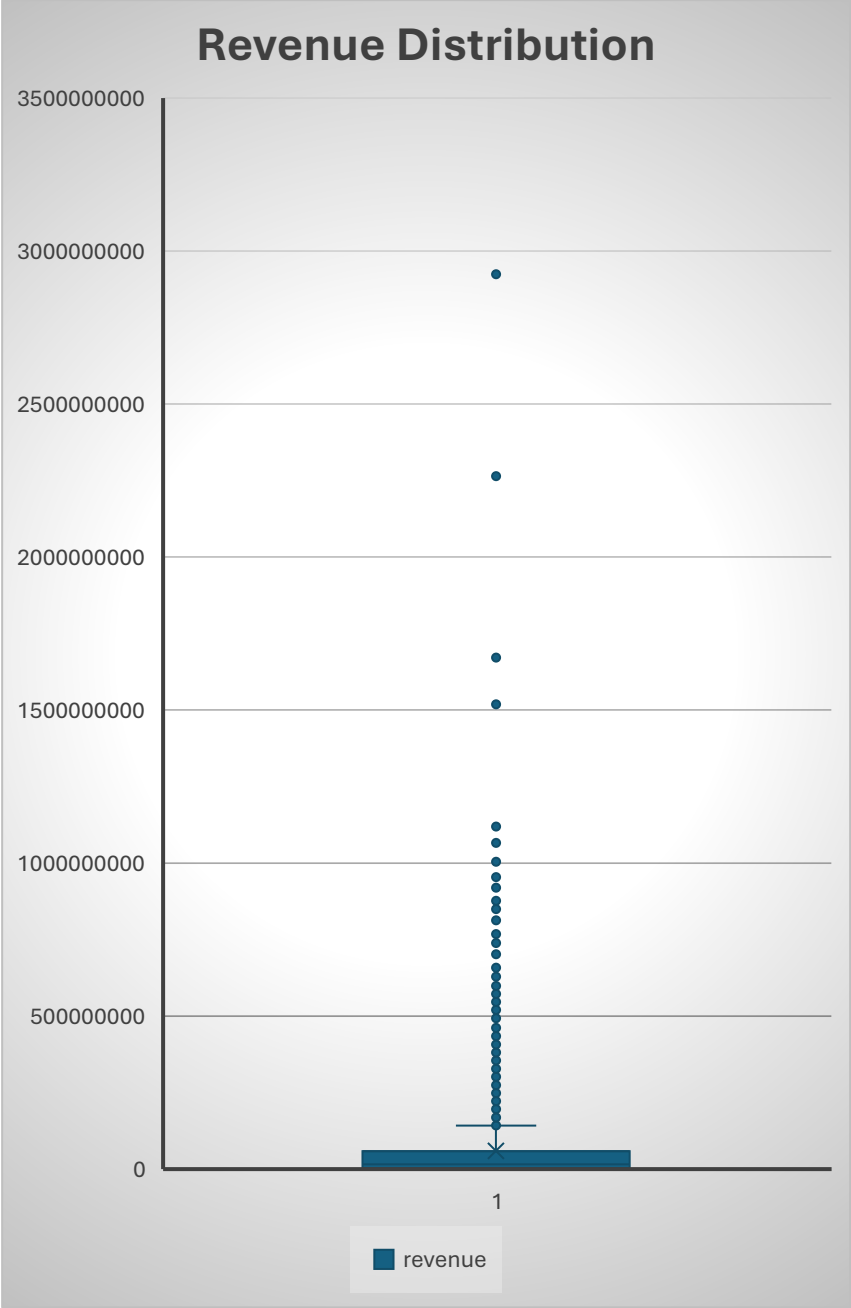
4. Budget vs Revenue (Scatter Plot)

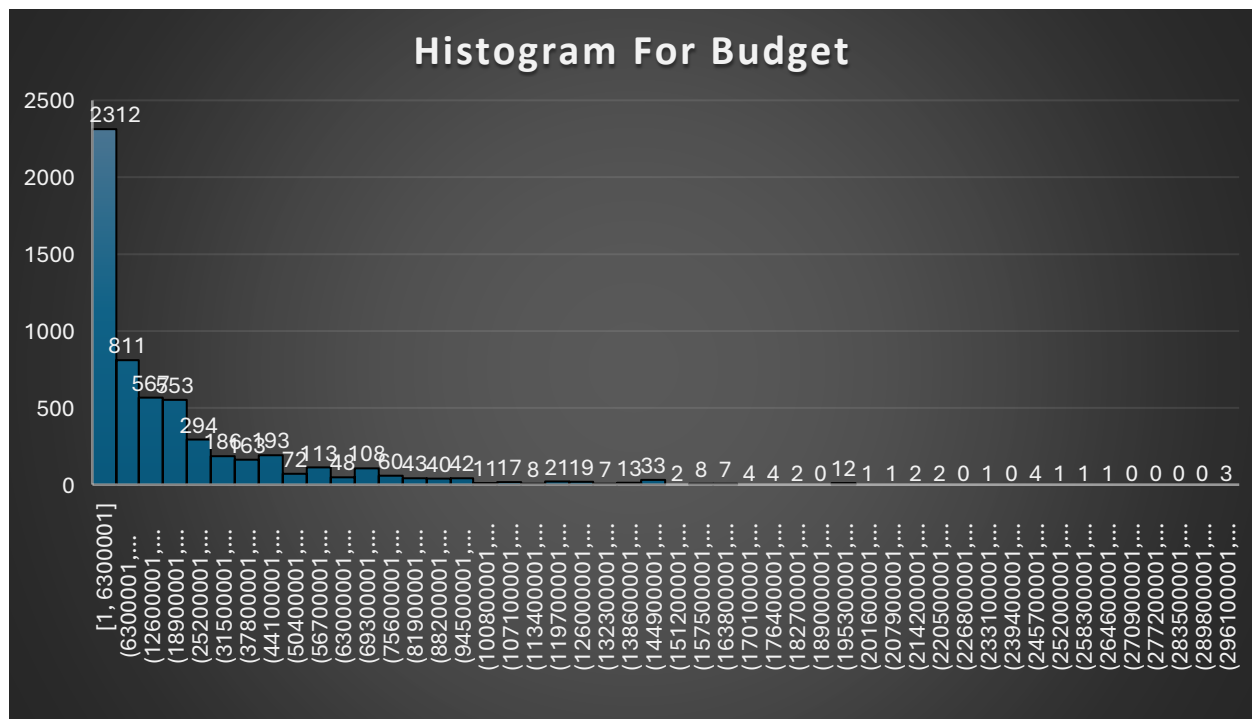
- Moderate - Strong positive correlation found ($r \approx 0.69$, $R^2 \approx 0.49$).
- Indicates that higher budget generally leads to higher revenue but not always profit.



5. Revenue Distribution (Histogram & Box Plot)

- Skewed right with outliers (e.g., blockbuster films). Therefore, we can not remove these as it is not incorrect data i.e some films earn many more
- However, Many films operate under \$50M revenue threshold.





6. Standard Deviation

- STDEV.P was used to understand variability:
 - Revenue SD: High → Indicates diverse outcomes
 - Budget SD: Moderate

Standard Deviation - Revenue		Standard Deviation - Budget
Measure variability in film earnings		Check how consistent film budgets are
126435255		32425385.76

7. Correlation Heatmap

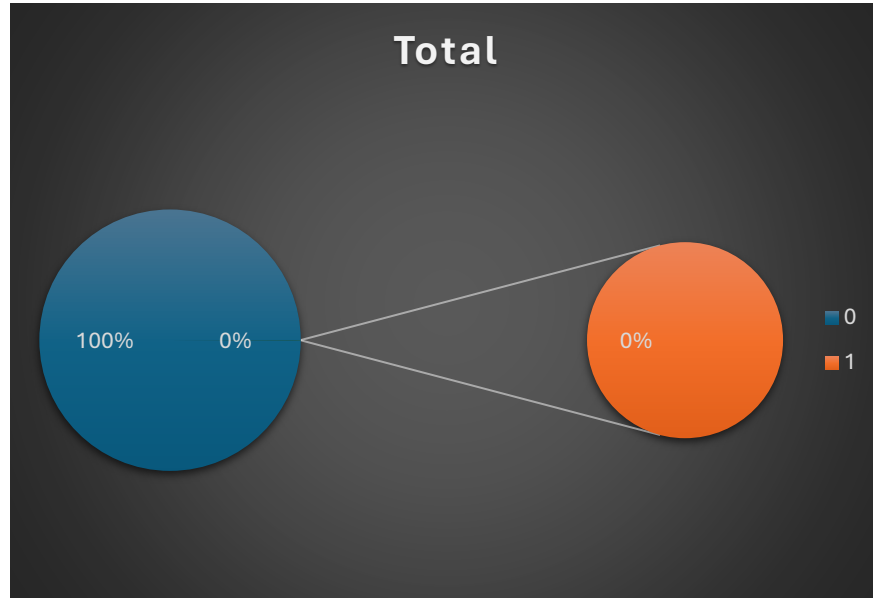
- Created with conditional formatting to show relationships:

	budget	revenue	Avgratings	runtime
budget	1	0.69	0	0.21
revenue	0.69	1	0.18	0.2
Avgratings	0	0.18	1	0.31
runtime	0.21	0.2	0.31	1

Additional Insights

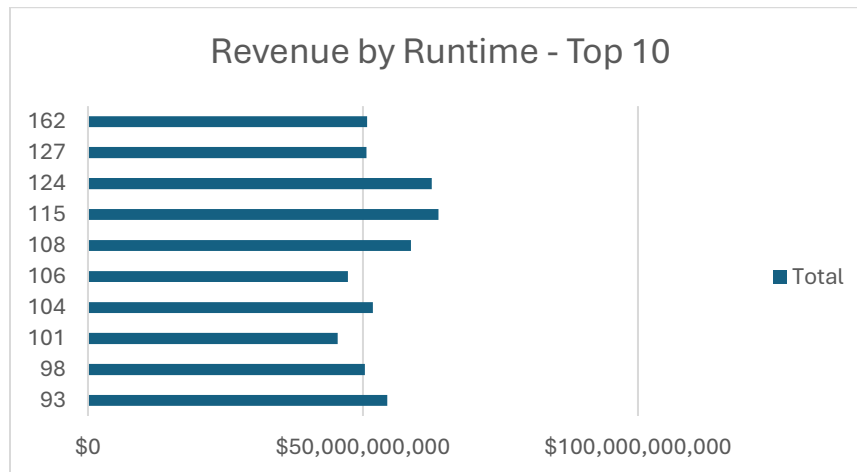
- **IsAdult vs Revenue:**

- Adult films had lower average revenue.



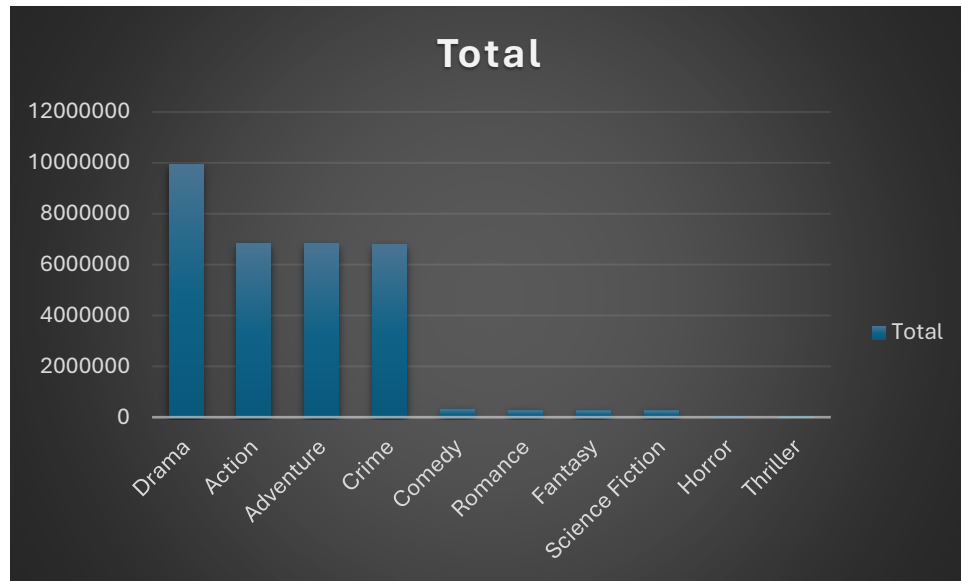
- **Runtime vs Revenue:**

- On general , longer films tend to generate more revenue.



- **High ROI Cases:**

- Interestingly, the highest ROI by Genre is not the same as Highest Revenue By Genre



Dashboard: Financial Movie Analysis

